

Do Historical Moments Improve the Risk–Work Frontier?

A Controlled Study of Scheduled Quadratic Network Growth

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Abstract

Choosing a promising direction for a new neural unit does not establish an advantage after finite optimization and charged proposal costs. We study this distinction in a centered quadratic network whose hidden vectors, signed output coefficients, and intercept are all trained. A historical Gaussian moment proposes spectral directions while retaining 128 examples; the comparators use random growth with 128 or 138 examples, or a fixed width of seven with 138 examples. Growth from one to seven units follows a fixed schedule. A locally locked protocol specifies five work budgets, balanced learning-rate selection on separate development problems, and 256 independent confirmation teachers and streams. Exact population excess risk is evaluated only after each training trajectory is fixed. The primary endpoint averages terminal excess risk over a fixed log-budget grid and requires superiority to all three comparators. The prespecified conjunction is met ($p = 0.00312$): mean integrated excess risk is 0.00536, compared with 0.00594, 0.00579, and 0.01498 for the three controls. The fixed-width mean is driven by large-error runs; historical moments win on only 118/256 paired seeds. The effect is modest against random growth, and both random controls already meet the fixed risk target on every seed at the smallest evaluated budget. Fixed width is faster in the observed learner timings. Secondary tasks do not show a uniform historical-moment advantage. The analytical identities and empirical comparison concern this synthetic model and optimizer; they supply neither autonomous growth nor a general risk or runtime guarantee.

1 Question and relation to earlier work

A network can be enlarged without changing its current output. It can also receive a direction that would reduce risk under an ideal coefficient refit. The practical question is whether this information improves the predictor that training actually reaches under a common resource rule. Direction estimation, replay maintenance, optimizer state, and fitting all consume resources. An initially better direction can fail to repay these costs, and a competent fixed-width model may reach the same risk without any growth mechanism.

This paper studies one narrowly specified version of that question: does retaining a historical quadratic moment reduce final population excess risk over a prespecified training-work grid, compared with two random-growth controls and a fixed-width control? Earlier internal Phase 5 experiments did not establish that stronger birth directions produced lower final risk. Phase 6 therefore evaluates a complete budget frontier using independently rotated teachers, balanced development selection, and broader paired replication. Its intended contribution is a controlled empirical finding and an inspectable resource ledger. The historical-moment mechanism is carried forward as a candidate to test, not introduced as a new algorithmic principle.

The relevant prior work already separates several ingredients of this construction. Net2Net develops function-preserving transformations for transferring a trained network into wider or deeper networks [1]. GradMax chooses initializations for added neurons by maximizing new-weight gradients and uses a singular-value decomposition [2]. Growing Tiny Networks formalizes expressivity bottlenecks relative to functional gradients and uses this information to add neurons during training [4]. In the matrix formulation, GECCO is a greedy method for convex optimization under a low-rank constraint, using leading singular directions and corrective optimization [3]. These are close precedents for function preservation, gradient-informed expansion, and spectral rank-one search. We do not claim that these ingredients are new.

The present factorized quadratic model permits exact population evaluation and transparent small-scale accounting. It is not a reproduction of those prior algorithms, and the four experimental comparators do not establish superiority over them. In particular, finite clipped Adam updates of a nonlinear parameterization

differ from an analytical rank-one coefficient fit or a fully corrective matrix optimization step. Section 2 gives the exact risk identity and states this separation. The earlier project also considered a fresh-block matrix recurrence; the discussion there is included solely to explain why its contraction analysis does not transfer to this implementation.

2 Mathematical scope and the budget endpoint

The mathematical ingredients below are classical Gaussian moment identities and elementary properties of the chosen evaluation endpoint. They are not claimed as novel learning principles. Greedy low-rank optimization and growth based on expressivity bottlenecks already have established antecedents [3, 4]. Our question is empirical: does the information supplied by a historical moment improve the training-budget frontier once its computational costs are charged?

2.1 Exact population risk of the trained predictor

Let $X \sim N(0, I_d)$, $H(X) = XX^\top - I_d$, and $q_A(X) = \langle A, H(X) \rangle$, where matrices are symmetric and $\langle A, B \rangle = \text{tr}(AB)$. The teacher and predictor are

$$Y = q_{A_*}(X) + c_* + \xi, \quad f_{B,c}(X) = q_B(X) + c, \quad B = \sum_{j=1}^k a_j w_j w_j^\top, \quad (1)$$

where ξ is independent of X , with mean zero and variance $\sigma^2 < \infty$. Our experiments set $c_* = 0$ but fit c .

Proposition 1 (Exact risk including the intercept). *For every symmetric B and scalar c ,*

$$R(B, c) := \mathbb{E}[(Y - f_{B,c}(X))^2] = \sigma^2 + 2\|B - A_*\|_F^2 + (c - c_*)^2. \quad (2)$$

Consequently, the excess risk used in the experiments is $\mathcal{E}(B, c) = 2\|B - A_\|_F^2 + c^2$. The same formula holds conditionally on a realized training history and teacher, provided the test example is independent of that history.*

Proof. The Gaussian second- and fourth-moment identities give

$$\mathbb{E}[X_i X_j] = \delta_{ij}, \quad \mathbb{E}[X_i X_j X_k X_l] = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}.$$

Thus $\mathbb{E}H_{ij} = 0$ and $\mathbb{E}[H_{ij}H_{kl}] = \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$. Multiplying by $D_{ij}D_{kl}$ and summing gives $\mathbb{E}[q_D(X)^2] = 2\|D\|_F^2$ for symmetric D . Take $D = A_* - B$. Since $\mathbb{E}q_D = 0$, the cross term between q_D and $c_* - c$ vanishes. Noise independence and zero mean eliminate its cross terms with both quantities. Expanding $(q_D + c_* - c + \xi)^2$ proves (2). Conditioning on a saved predictor and its training history fixes B, c ; the independent test input retains the same Gaussian moments. $\square$

The intercept term cannot be omitted: $B = A_*$ and $c = 1$, for example, have excess risk one when $c_* = 0$. For each individual run, a separate evaluator applies (2) after its training trajectory has been fixed. Development and confirmation have distinct roles. Before observing any new development results, we froze a common search protocol: all four methods are evaluated at learning rates $\{0.0075, 0.015, 0.03\}$ and all five budgets on 16 independent development teacher/data seeds. For each method, the mean exact population AUC across these development seeds selects one learning rate, which is then frozen before the 256-replicate confirmation experiment. This is explicitly an oracle-based development selection; its computational cost is reported separately. Confirmation teacher matrices, population scores and oracle direction coefficients do not enter the learner’s proposal selection, optimization, stopping, or hyperparameter selection. Exact evaluation removes test-input Monte Carlo error in this synthetic model; it does not remove variation across independently generated training problems.

2.2 Function-preserving birth and the limits of direction quality

Appending any finite direction v with output coefficient $a_{k+1} = 0$ leaves both B and c unchanged, and hence preserves the predictor for every input. Initializing the new Adam moment states to zero also leaves the function unchanged. The subsequent optimization trajectory is a separate matter: at birth the gradient with respect to the new hidden vector is zero because it is multiplied by its output coefficient, while the new output coefficient can have a nonzero gradient.

For a unit direction v , set $E = A_* - B$ and hold all other parameters, including the intercept, fixed. Proposition 1 yields

$$R(B + \alpha vv^\top, c) - R(B, c) = 2\alpha^2 - 4\alpha v^\top E v. \quad (3)$$

The oracle scalar coefficient is $\alpha = v^\top E v$, with improvement $2(v^\top E v)^2$. Maximizing this quantity over unit directions gives $2\|E\|_{\text{op}}^2$. This is the improvement available from an ideal scalar refit at a fixed predictor, not an improvement guaranteed by the implemented factor Adam updates.

In particular, the earlier project’s fresh-block quadratic analysis concerns a prescribed update of the form

$$B^+ = B + \frac{1}{2n} \sum_{i=1}^n \{Y_i - q_B(X_i)\} H(X_i),$$

under its stated zero-intercept model and conditional independence assumptions. Factor Adam instead updates (w_j, a_j, c) using empirical gradients, accumulated coordinatewise moments, and clipping. It may reuse retained examples and historical summaries. Neither its update map nor its sample dependence matches that matrix recurrence, so the contraction theorem does not supply a convergence rate or cost bound for the present experiment. Even the exact empirical residual matrix, when the predictor has an intercept, is

$$\begin{aligned} G_n(B, c) &= S_n - T_n(B) - \tfrac{1}{2}c\bar{H}_n, & S_n &= \frac{1}{2n} \sum_i Y_i H_i, \\ T_n(B) &= \frac{1}{2n} \sum_i q_B(X_i) H_i, & \bar{H}_n &= \frac{1}{n} \sum_i H_i. \end{aligned}$$

The historical proxy $S_n - B$ uses the known population Gaussian operator; it is not this exact empirical residual.

2.3 A fixed log-budget area from five completed runs

The five preregistered per-arrival budget caps, in the declared work units, are

$$(b_1, b_2, b_3, b_4, b_5) = 10^6(3.2, 5, 6.4, 8, 10).$$

Every run processes seven arrivals under the same data-availability rule. Let e_{mrj} be the final excess risk of method m , replicate r , and budget setting b_j , after all seven arrivals. Each entry is obtained from its own completed training run. These five entries are not checkpoints extracted from one run with the largest budget.

Write $x_j = \log b_j$ and $L = x_5 - x_1 > 0$. Define the normalized log-budget trapezoid area

$$\text{AUC}_{mr} = \frac{1}{L} \sum_{j=1}^4 (x_{j+1} - x_j) \frac{e_{mrj} + e_{mr,j+1}}{2} = \sum_{j=1}^5 \omega_j e_{mrj}, \quad (4)$$

where

$$\omega_1 = \frac{x_2 - x_1}{2L}, \quad \omega_5 = \frac{x_5 - x_4}{2L}, \quad \omega_j = \frac{x_{j+1} - x_{j-1}}{2L} \quad (2 \leq j \leq 4).$$

The weights are positive, sum to one, and are fixed independently of all outcomes. Only budget is logarithmically transformed; the ordinate is excess risk itself. Lower area is better. Replacing b_j by its seven-arrival cap $7b_j$ leaves all weights unchanged, as does changing the logarithm base.

Equation (4) is an integral of the piecewise-linear interpolant in log budget, not an assumption that the unobserved performance curve is linear. The budget axis is the assigned cap; actual charged work, unused

terminal budget and wall time are reported separately. The area therefore summarizes the specified allocation protocol and cost ledger, and does not by itself establish hardware runtime dominance. The primary sample comprises 256 independent teacher/data replicates, with pairing across methods and budget settings within each replicate. The five budgets do not multiply the number of independent inference units by five.

2.4 Meaning of the three-comparison intersection–union test

Let h denote historical-moment growth, and let m_1, m_2, m_3 denote the three preregistered primary comparators. For $\ell = 1, 2, 3$, define the paired area difference and its mean over the specified distribution of independently generated problems by

$$D_{r\ell} = \text{AUC}_{hr} - \text{AUC}_{m_{\ell}r}, \quad \mu_{\ell} = \mathbb{E}[D_{r\ell}].$$

The three-comparison intersection–union test (IUT3) addresses

$$H_0 : \bigcup_{\ell=1}^3 \{\mu_{\ell} \geq 0\}, \quad H_1 : \bigcap_{\ell=1}^3 \{\mu_{\ell} < 0\}. \quad (5)$$

If p_{ℓ} is a valid one-sided p -value for the corresponding component null, reject (5) at level α only when all three $p_{\ell} \leq \alpha$, equivalently when $p_{\text{IUT}} = \max_{\ell} p_{\ell} \leq \alpha$. Indeed, under the union null at least one component null, say ℓ_0 , is true, and

$$\mathbb{P}(p_{\text{IUT}} \leq \alpha) \leq \mathbb{P}(p_{\ell_0} \leq \alpha) \leq \alpha.$$

This argument requires no independence between the three comparisons and no Bonferroni division of α for this single conjunction claim. Its calibration still depends on the validity of each chosen component test; a finite-sample exact claim cannot be inferred merely from having 256 replicates. Additional families of claims or outcome-dependent selection of comparators require their own handling.

Rejection supports lower mean area against every one of the three specified comparators under this protocol. It does not imply lower risk at every budget, success on every seed, superiority to untested methods, or optimality among architectures. Failure to reject does not establish equivalence, absence of a directional effect, or absence of an advantage at some budgets. A superiority test has no prespecified equivalence margin. Together, the risk identity, direction diagnostic and IUT3 interpretation provide no uniform optimal cost bound for factor Adam or autonomous network growth.

3 Experimental protocol

3.1 Data stream and learner information

Each primary replicate uses $d = 12$ and seven successive arrivals of 1,024 independent observations, for 7,168 observations in total. The teacher is $A_* = Q\Lambda Q^{\top}$, where Q is obtained from an independent Gaussian QR draw and

$$\text{diag}(\Lambda) = (0.65, -0.50, 0.30, -0.20, 0, \dots, 0).$$

Inputs are independent $N(0, I_{12})$, the teacher intercept is zero, and additive Gaussian noise has standard deviation 0.3. Each replicate has its own teacher rotation, observations, noise, and optimization random streams. Methods and budgets are paired within the same replicate. At each arrival, a method accesses only the current block and its permitted retained state; it never accesses future observations and never receives the teacher matrix or a population score while training.

At an arrival, the fitting pool consists of the fresh 1,024 observations and the reservoir retained from previous arrivals. A reservoir update then processes the fresh block using Algorithm R. The fresh observations are not duplicated in that arrival’s fitting pool merely because some of them enter the updated reservoir. Independent random namespaces separate data generation, initialization, proposal directions, fitting permutations, reservoir selection, and execution order. Within a seed, all trajectories are completed before a separate evaluator adds their exact population risks. The evaluator’s access to A_* is an evaluation permission, not a learner input.

Method	Width	Birth direction	Replay rows	Retained bytes
Historical moment	1 $\rightarrow$ 7	Historical spectral	128	15,488
Random R128	1 $\rightarrow$ 7	Random	128	14,336
Random R138	1 $\rightarrow$ 7	Random	138	15,456
Fixed width 7	7	No births	138	15,456

Table 1: Persistent observation-dependent arrays after filling the reservoir. Every stored row has 12 float64 inputs, a float64 label, and an int64 observation identifier. The historical matrix adds 144 float64 words. Its 32-byte difference from the 138-row controls is disclosed; storage is close but not exactly matched. Model, optimizer, working arrays, and control state are separate.

3.2 Four complete algorithms

The prediction function implemented in all four methods is

$$f_{\theta}(x) = c + \sum_{j=1}^k a_j ((w_j^{\top} x)^2 - \|w_j\|_2^2). \quad (6)$$

All w_j , a_j , and c are trainable; the comparison is not a fixed-feature regression. At initialization, hidden vectors are independent Gaussian draws normalized to unit length, output coefficients are drawn from $N(0, 0.01^2)$, and $c = 0$. The three growing methods start at width one. Before fitting arrivals two through seven, each appends exactly one hidden unit. Its output coefficient and Adam moments are zero, while existing parameters and states are preserved. The fixed-width method initializes seven units and optimizes all of them from the first arrival.

Historical moment retains a full 12×12 matrix sum and a reservoir of 128 examples. After incorporating the fresh block, let N_t denote the number of observations received and define

$$S_t = \frac{1}{2N_t} \sum_{i \leq N_t} Y_i (X_i X_i^{\top} - I_d), \quad M_t = S_t - B_{t-}. \quad (7)$$

The stored sum is updated once per observation; no historical raw data beyond the reservoir are retained by the learner. The new direction is a unit eigenvector of M_t associated with its largest absolute eigenvalue. Both signs of teacher eigenvalues matter because the output coefficients are signed. The Gaussian population identity $\mathbb{E}S_t = A_*$ motivates this proxy, but M_t is not the exact empirical residual matrix of the fitted predictor with intercept.

Random R128 and Random R138 use normalized independent Gaussian birth directions and retain 128 and 138 examples, respectively. The former controls for the number of replay examples; the latter spends approximately the historical matrix’s storage on additional examples. Fixed width 7 retains 138 examples. No method uses a gate, a validation controller, an accepted/rejected proposal branch, or an autonomous choice of birth timing or width.

3.3 Fitting and optimizer-age sensitivity

The empirical squared-error gradients update all parameters using Adam with $(\beta_1, \beta_2) = (0.9, 0.999)$, numerical stabilizer 10^{-8} , and global gradient-norm clipping at 20. The batch size is 128. Random permutations are traversed with tails carried into the next permutation, so an epoch boundary alone does not create a short optimizer batch. If the remaining budget cannot fund 128 examples, the largest affordable positive final batch is used and recorded. Training stops when no positive batch fits the remaining declared work. There is no validation-based early stopping or rollback.

The primary optimizer uses the global Adam update count for bias correction, with zero moment states for newborn coordinates. This choice gives new and old coordinates different histories despite a shared global count. A separately specified sensitivity uses each coordinate’s local age for bias correction; newborn ages start at zero and existing ages continue. The sensitivity repeats the first 32 primary seeds with frozen learning rates. Its results are paired with those same seeds’ primary runs and are not 32 additional independent primary replicates.

3.4 Allocated budgets and charged operations

Each method runs independently at each of the five per-arrival caps $10^6(3.2, 5, 6.4, 8, 10)$. Thus the full-stream allocated totals are $10^6(22.4, 35, 44.8, 56, 70)$. A higher-budget run does not continue a lower-budget run. Before fitting each arrival, the implementation reserves declared costs for initialization when applicable, observation identifiers, retention allocation and maintenance, fitting-pool copies, historical-moment maintenance, residual-matrix construction, eigendecomposition, direction construction, births, and event control. The remaining allocation funds fitting and permutation operations.

Work is a deterministic scalar-operation proxy defined by explicit formulas in the runner. It is not measured hardware FLOPs. For a batch of n examples at width k , let $P = (d + 1)k + 1$. The fitting charge is

$$C_{\text{fit}}(n, d, k) = n(6dk + 12k + 2d + 12) + 30P + 4dk + 4k + 32. \quad (8)$$

Each newly generated permutation of a fitting pool of size L costs $4L$. A moment update costs $2nd^2 + nd + n + 4d^2$, and an eigendecomposition costs the declared surrogate $10d^3$. Additional formulas and the separation of measurement categories are given in Appendix A. All methods pay their own charges under the same definitions, so equal allocated work need not yield equal optimizer steps. Every arrival records actual charged work and its integer remainder.

Learner timings sum algorithm blocks covering initialization, proposal, retention, optimization, and cleanup. Data generation, post-trajectory evaluation, archive serialization, and major audit-counter updates are excluded from those blocks; minor control, allocation, and instrumentation overhead remains. The code inventories retained arrays, parameter and optimizer arrays, and estimated explicit working arrays. These are neither operating-system peak resident memory nor a measurement of NumPy/BLAS internal allocations. Wall times depend on the implementation, process contention, memory allocation, and the host; a declared-work advantage is not automatically a wall-clock advantage.

3.5 Separate development selection and primary inference

Before confirmation, every method receives the same selection rule: 16 development seeds, all five budgets, and three learning rates, $\{0.0075, 0.015, 0.03\}$. The method-specific rate minimizing mean development AUC is selected, with the smaller rate breaking an exact tie. This entails 960 complete development trajectories. Development uses independently generated teachers and streams, and its exact population scores are an explicitly disclosed selection oracle. The selected rates are frozen for confirmation, secondary tasks, and the local-age sensitivity. Development work is a one-time experimental selection cost, disclosed separately from a single trained instance’s work.

The primary confirmation sample comprises exactly 256 seeds. For each comparator we compute the seedwise difference in the AUC defined in Equation (4). The primary decision requires all three one-sided paired- t upper 95% confidence bounds to lie below zero. This is the intersection–union test of Equation (5). We report its maximum component p -value, each paired standard deviation and interval, and 10,000-resample paired bootstrap intervals as a sensitivity check. The t approximation is not a finite-sample guarantee for unbounded squared loss. A conservative planning calculation at $N = 256$ requires roughly a 0.20 standardized effect in the weakest contrast for 80% joint power. This is a planning approximation, not retrospective evidence of power or equivalence.

The protocol was locally locked before new smoke or development outcomes, with a subsequent code freeze following correctness review. It was not externally registered or timestamp-certified. The seeds, budgets, primary statistic, learning-rate candidates, and sample size were fixed prospectively. Smoke runs serve correctness and timing checks only. Necessary implementation corrections and their source snapshots are preserved. Poor-performing seeds remain in the sample; nonfinite or incomplete runs require an explicit investigation instead of silent exclusion. Confirmation results do not authorize an additional seed, budget, task, or optimizer search.

3.6 Secondary questions

The fixed target for the primary noisy task is excess risk $\epsilon = 0.01$. We report attainment at every budget and the first successful evaluated final-budget point, retaining failures as nonattainment. This is a discrete

Comparator	Mean diff.	One-sided upper 95%	p	Bootstrap 95% CI
Random R128	-0.00058145	-0.00051507	2.593e-35	[-0.00066027, -0.00050225]
Random R138	-0.00043108	-0.00033745	2.823e-13	[-0.00053926, -0.00031965]
Fixed width 7	-0.0096221	-0.0038628	0.003117	[-0.017273, -0.0038145]

Table 2: Primary normalized excess-risk/log-allocated-work AUC differences (historical moment minus comparator), paired over 256 independent teachers and streams. Negative values favor historical moments. Bootstrap intervals use 10,000 paired resamples and are sensitivity analyses only.

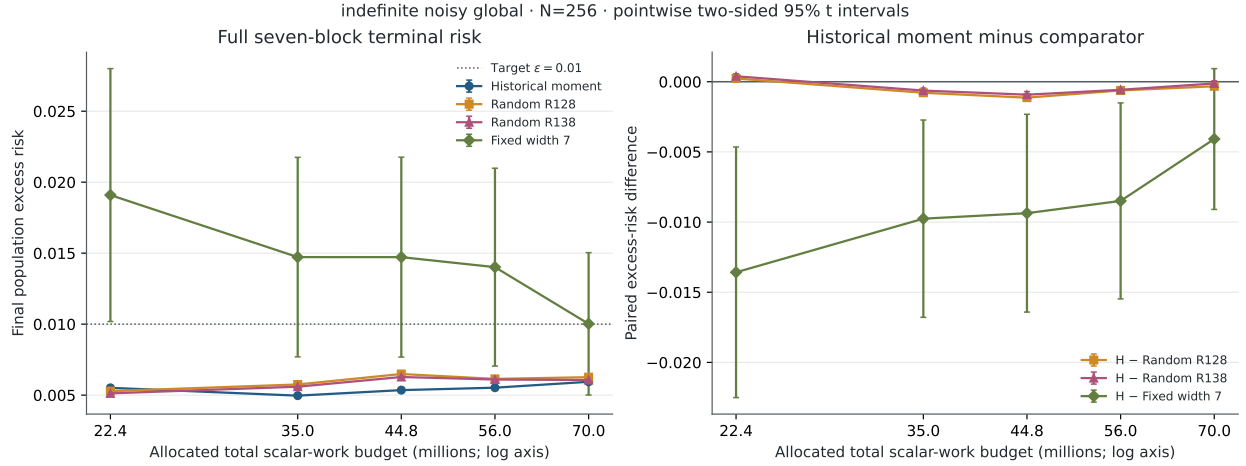


Figure 1: Primary final-risk frontier and seed-paired differences after seven arrivals. Each budget setting is trained independently for every seed; plotted points are means over 256 seeds. Lines connect evaluated budget settings. Error bars are descriptive 95% intervals across the 256 paired problems, not simultaneous confidence bands. Only the allocated-work axis is logarithmic.

five-point grid summary. It does not assume a monotone risk curve, interpolate an unmeasured crossing, or estimate the earliest training-time crossing.

Two predeclared compression comparisons place the historical method at 80% of the comparator’s allocated work: per-arrival 6.4 versus 8 million units and 8 versus 10 million units. We report paired terminal-risk differences and target attainment. These exploratory comparisons do not establish a unique 20% savings, because a comparator might also reach the target at a lower budget.

Two descriptive controls each use 32 independent seeds and all four methods at all five budgets. The null teacher has $A_* = 0$ and noise standard deviation 0.5; it probes scheduled expansion when no quadratic signal is present. The noiseless rank-two teacher has eigenvalues 0.8 and 0.45 with independent rotation; it probes dependence on label noise and spectrum. Neither control supplies additional primary-test replications.

4 Results

4.1 Frozen selection and primary frontier

The prespecified conjunction met the primary criterion (intersection-union $p = 0.0031173$; maximum one-sided upper bound -0.00033745). The endpoint integrates raw excess risk over the fixed log allocated-work grid; it is not a work-reduction percentage. Pointwise intervals and compressed-budget comparisons are descriptive. Development selected the following learning rates: Historical moment: 0.0075, Random R128: 0.0075, Random R138: 0.0075, Fixed width 7: 0.0075. All development trials consumed 43,812,371,616 scalar-work proxy units and 75.075 summed learner seconds. These costs are disclosed separately from per-instance deployment work.

Mean primary AUC is 0.0053573 for historical moments, 0.0059388 for R128, 0.0057884 for R138, and

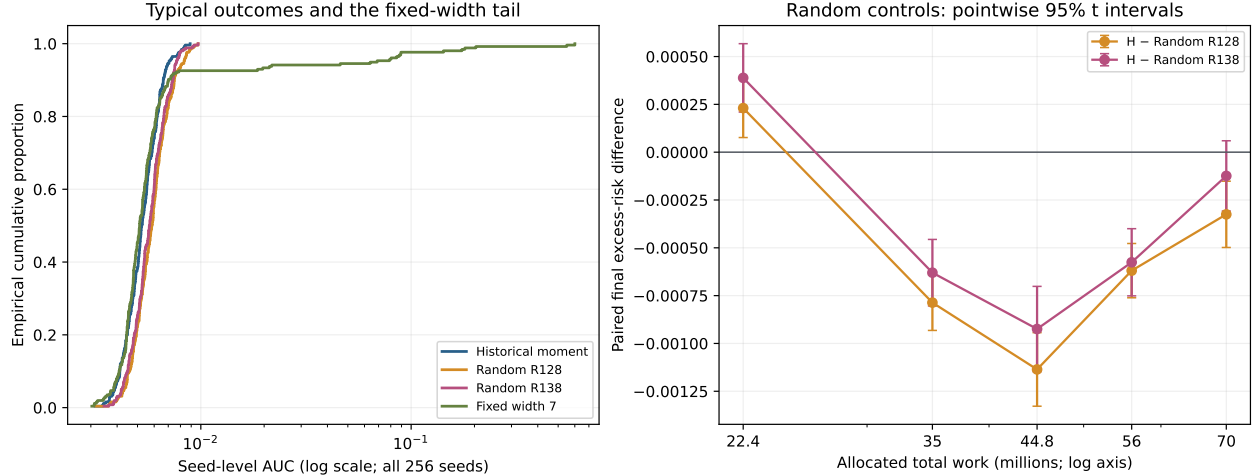


Figure 2: Descriptive post-analysis visualization of all 256 primary seeds. Left: empirical cumulative distributions of seed-level AUC, including every large fixed-width value. The logarithmic display axis does not change the raw-excess-risk primary endpoint. Right: historical minus random-growth terminal-risk differences at each allocated budget, with ordinary paired 95% t intervals. Positive differences at the smallest budget favor the random controls; these pointwise intervals are not simultaneous confidence bands.

0.014979 for fixed width seven. The historical reductions relative to the two random controls are about 9.8% and 7.4% of their *excess-risk areas*. They are not reductions of total risk, allocated work, or elapsed time. The paired standard deviations are 0.0006434, 0.0009075, and 0.05582, respectively. The fixed-width contrast is therefore much more variable and tail-sensitive than the random-growth contrasts. A descriptive audit of its distribution shows median AUC 0.005112, slightly below the historical median 0.005237, and maximum AUC 0.60066. Historical moments have lower paired AUC on 118/256 problems and higher AUC on 138; the median paired difference is +0.00008489. The five problems most favorable to historical moments account for 68.34% of the net mean difference. Thus this contrast concerns *expected mean risk and large-error outcomes*, not a typical-seed or majority-seed advantage. These post-analysis distribution diagnostics do not alter the primary endpoint or exclude any seed. The paired bootstrap interval remains negative, but it resamples only the observed tail. Neither that interval nor the t approximation supplies distribution-free finite-sample coverage or resolves uncertainty about unobserved large errors.

The average-area result does not imply pointwise dominance. At the smallest cap, historical moments have mean excess risk 0.005514, versus 0.005284 and 0.005126 for R128 and R138. The historical method has a lower mean than both random controls at the other four caps. Its mean risk is lowest at the 5-million per-arrival cap and then rises, reaching 0.005940 at 10 million. The observed curves are not monotone training-efficiency frontiers. They describe terminal performance under fixed-rate fitting and a fixed arrival schedule; there is no selected best-checkpoint envelope.

Figure 2 makes the fixed-width upper tail and the smallest-budget reversal against random growth visible using all 256 primary seeds. This descriptive post-analysis visualization adds no hypothesis test and excludes no observation.

All methods selected 0.0075, the smallest candidate learning rate. Thus balanced selection was completed as planned, but the development grid does not establish globally optimized controls or rule out improvement from lower rates. The locked confirmation was completed without extending that grid after observing the result.

4.2 Target attainment, charged resources, and sensitivity

At the smallest evaluated cap, the 0.01 target is attained by 255/256 historical runs, 256/256 R128 runs, 256/256 R138 runs, and 237/256 fixed-width runs. Across the complete evaluated grid, the corresponding counts are 256, 256, 256, and 248; eight fixed-width problems never attain it. First attainment is already

Task / age (N)	Historical	R128	R138	Fixed 7
Primary / global (256)	0.005357	0.005939	0.005788	0.014979
Null / global (32)	0.008164	0.009189	0.008923	0.007243
Rank two / global (32)	1.142×10^{-6}	7.639×10^{-7}	8.701×10^{-7}	1.091×10^{-4}
Primary / local (32)	0.004689	0.005483	0.005675	0.024995

Table 3: Mean normalized terminal excess-risk areas. Only the first row belongs to the primary inference. The local-age row reuses the first 32 primary problems; it is not a comparison with all 256 global-age runs. Full intervals and paired contrasts are retained in the analysis CSVs and generated detailed table.

Method	Mean charged work (10^6)	Mean Adam steps	Mean seconds
Historical moment	45.63859	1,144.0	0.08494
Random R128	45.63651	1,203.2	0.08840
Random R138	45.63806	1,203.4	0.08840
Fixed width 7	45.63838	547.4	0.04790

Table 4: Primary-run resources, averaged equally over 256 seeds and five independent budget settings per method. Mean allocated work is 45.64 million units. Seconds sum timed learner blocks for one complete trajectory; they are implementation observations under concurrent execution, not a controlled hardware benchmark.

left-censored by the grid for nearly all growing runs, so the target cannot identify a historical-moment work saving. After their first successful grid point, 3 historical, 16 R128, and 19 R138 problems fail at a later point. This supports reporting every evaluated outcome instead of assuming monotonicity.

The 6.4-versus-8-million compression comparison favors historical moments in mean excess risk against R128, R138, and fixed width: the differences are -0.000791 , -0.000747 , and -0.008663 , with descriptive 95% paired intervals $[-0.000948, -0.000634]$, $[-0.000929, -0.000566]$, and $[-0.015640, -0.001687]$. In the 8-versus-10-million comparison, the differences are -0.000740 , -0.000539 , and -0.004498 ; the fixed-width interval $[-0.009526, 0.000529]$ crosses zero. These comparisons use approximately 0.8000 times the charged work, but do not demonstrate 20% minimum-cost savings. In particular, both random controls already reach the target at the lowest cap.

On the same 32 primary problems, replacing global by local ages reduces the historical method’s mean AUC by 0.0006833 (descriptive 95% paired interval $[-0.0008481, -0.0005184]$). Fixed-width trajectories are unchanged. Under local age, historical minus R128, R138, and fixed-width mean differences remain negative, at -0.0007941 , -0.0009863 , and -0.02031 ; their ordinary two-sided t intervals cross zero for R138 and fixed width. The direction of the means survives this sensitivity, but its smaller sample does not establish the same three-way precision. It is not used to replace or improve the prespecified primary decision.

The task controls show the scope of the result. With the null teacher, historical minus fixed-width AUC is 0.0009212 ($[-0.0001335, 0.001976]$). With the noiseless rank-two teacher, historical AUC exceeds R128 by 3.783×10^{-7} ($[1.569 \times 10^{-8}, 7.409 \times 10^{-7}]$), while all growing methods reach very small absolute errors. Thus the positive primary result is not a uniform ranking across even these two nearby synthetic tasks. These intervals are descriptive.

The primary study charges 2.33666×10^{11} work units across 5,120 trajectories, against 2.336768×10^{11} allocated. Total unused work per trajectory ranges from 413 to 7,269 units; every arrival respects its cap. The historical method performs fewer Adam steps than random growth because its summary and proposal charges leave less work for fitting. The comparison therefore concerns complete algorithms, not an isolated intervention on the birth direction. Summed primary learner time is 396.35 seconds, while elapsed invocation time is 215.19 seconds under concurrent execution. Fixed width is faster in these observed timings despite its higher primary mean risk, so the result does not establish wall-clock dominance. Secondary and age-sensitivity invocations add 54.55 and 29.55 elapsed seconds, respectively, with their costs retained separately.

5 Discussion and limits

The locked study detects a narrow positive finding: historical-moment growth has lower mean terminal excess-risk area than all three specified controls on the indefinite noisy task. Its modest advantage over random growth remains after charging moment maintenance and search. The local-age means point in the same direction, with wider uncertainty. The work-target evidence and wall times do not establish a minimum-cost or general runtime advantage. These qualifications are part of the finding, rather than secondary exceptions to a broad efficiency claim.

The comparison with fixed width is specifically an expected-mean statement: fixed width has the lower paired AUC on a majority of primary seeds, while a few severe fixed-width errors dominate the mean difference. A reader valuing typical-case performance may therefore draw a different practical conclusion from the same complete records.

The experimental question concerns final predictors after all seven arrivals. A favorable newborn direction, a function-preserving birth, and a final-risk improvement are different statements. The first can be measured at a particular state; the second is an algebraic property; the third depends on the subsequent optimization and resource allocation. An analytical best-coefficient gain cannot replace the third measurement.

The two random controls also answer different questions. R128 holds the replay capacity fixed while changing direction information and the work needed to obtain it. R138 instead provides additional raw examples for approximately the same persistent data-summary storage. Neither comparison alone isolates every mechanism. Fixed width seven tests whether this scheduled growth procedure is useful relative to a full-width model trained from the start with a separately selected learning rate. Because earlier growing stages are narrower, their per-batch charges and optimizer update counts differ from fixed width. Those differences are part of the specified algorithms, rather than an unmeasured free benefit.

The primary endpoint is an average over a selected allocation grid. The positive finding supports lower mean integrated terminal excess risk against the three tested comparators under this protocol. It does not establish pointwise dominance, generalization to other spectra or input distributions, or a hardware-independent efficiency theorem. The local-age control helps expose sensitivity to the treatment of newborn coordinates, but it is smaller and uses rates selected under the global-age primary optimizer.

The known Gaussian input law is a substantial modeling advantage: it gives both the population operator behind the proposal and the exact risk evaluator. Non-Gaussian inputs, distribution drift, unmodeled nonlinear targets, and large-dimensional networks may change both. Memory scales quadratically with dimension for the historical matrix, whereas a retained row scales linearly. At $d = 12$ the full eigendecomposition is inexpensive enough to study locally; the declared $10d^3$ charge is not evidence about a large-scale spectral implementation. The three real neural-growth papers and GECCO cited above were not reimplemented as resource-matched baselines. A general comparative claim would require that additional work and an appropriate benchmark.

The bounded experiment is complete. Although the primary conjunction passes, the left-censored target comparison, faster fixed-width timings, nonmonotone budget curves, and task dependence do not justify promoting this finding to a broad efficiency claim. A detectable AUC difference alone does not establish a new learning principle. No additional search was undertaken to make the secondary outcomes more favorable, and intervals containing zero remain inconclusive for superiority rather than evidence of equivalence.

6 Reproducibility and artifact scope

The accompanying local artifact contains the prospectively specified protocol, frozen code, configuration and seed lists, generated teacher and observation arrays, per-arrival parameters and optimizer states, proposal records, data-access ledgers, raw results, independent audit reports, analysis tables, and vector figures. The paper’s numerical tables are generated from saved results rather than transcribed from plots. Code and configuration hashes connect each run to its implementation. The host is an Apple M1 Pro with eight logical CPUs and 16 GiB of memory; experiments use the existing NumPy CPU environment with BLAS thread counts pinned to one. Reproduction commands and exact environment metadata are provided with the artifact. The files constitute a local research record, not an external certification of execution or an already published study.

Independent saved-artifact checks passed for all 7,040 primary, secondary, and age-sensitivity trajectories: 49,280 arrival records and 112,640 saved parameter-risk evaluations were inspected, with maximum risk-

reconstruction discrepancy 1.11×10^{-15} . Separate checks cover analytical gradients, exact Gaussian quadrature, future-data isolation, unchanged non-timing trajectories under execution-order reversal, budget boundaries, optimizer ages, and deliberate artifact-corruption fixtures. Audits establish consistency of these saved records and implementation checks; they do not validate an untested statistical generalization. Earlier project files were protected by a 1,936-file SHA-256 baseline and remained unchanged.

References

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A Resource formulas and audit map

Table 5 gives the nonfitting charges used with Equation (8). Here $n = 1,024$, r is the previously retained count, C is reservoir capacity, and k^-, k are widths before and after birth. Each arrival’s charges respect its cap; unused remainders are recorded separately.

Operation	Declared scalar-work charge
Moment update	$2nd^2 + nd + n + 4d^2$
Historical residual matrix	$dk^- + d^2(2k^- - 1) + 2d^2$
Eigendecomposition	$10d^3$
Historical / random direction	$4d / 8d$
Reservoir maintenance	$n(d + 10)$
Fitting-pool copy	$(r + n)(d + 2)$
Arrival identifiers	n
Model initialization	$8dk + k + 5P$
Retention allocation	$C(d + 2)$, plus d^2 for a historical matrix
Birth and state copies	$4P + 5(d + 1)$
Event control	64
New fitting permutation	$4(r + n)$

Table 5: Accounting formulas charged when the corresponding operation occurs. Initialization and retention allocation occur only once; birth-related charges occur only at scheduled births. Reservoir maintenance conservatively charges each incoming observation, including those not retained.

The $P = (d + 1)k + 1$ parameters, Adam first and second moments, and coordinate ages occupy $32P$ bytes: 92 parameters and 2,944 bytes at width seven. Retained-data bytes appear in Table 1. Working arrays are estimated separately. Python and control state, vendor-internal allocations, data-owner arrays, and audit archives are excluded; this inventory does not establish peak-memory matching.